

Thus, by (2.2.3), the assertion will be proved if one shows that, for every $i \in [0, r]$, ξ^i is a free element of the A -module $H^{2i}(P, \mu^{\otimes i})$. It is enough to check this for ξ^r , which will be an immediate consequence of 2.2.2 c).

We now prove part c). In practice, this is the definition of the trace morphism in this situation; in any case, it expresses the compatibility of the trace morphism and the fundamental class in the diagram

$$\begin{array}{ccc} x_0 & \xrightarrow{\text{inj}} & \mathbb{P}_k^r \\ & \searrow & \downarrow p \\ & & k, \end{array}$$

where $\text{inj} : x_0 \rightarrow \mathbb{P}_k^r$ is the injection of a closed point of $\mathbb{P}_k^r$.

Corollary 2.2.4. Let L be a complex of A_S -modules. The arrow

$$\text{R}\Gamma(S, Y \overset{\text{L}}{\otimes}_A \text{id}(L)) : \bigoplus_{i=0}^r \text{R}\Gamma(S, L \otimes \mu_S^{\otimes -i})[-2i] \longrightarrow \text{R}\Gamma(P, p^*(L))$$

is an isomorphism.

Indeed, it is enough to apply the functor $\text{R}\Gamma_S$ to the isomorphism $Y \overset{\text{L}}{\otimes}_A \text{id}_L$.

2.2.5

To state the next corollary, for every scheme X we shall put

$$A^*(X) = \bigoplus_i H^{2i}(X, \mu_X^{\otimes i})$$

and shall use this notation systematically in what follows.

Corollary 2.2.6. The inverse-image homomorphism

$$p^* : A^*(S) = \bigoplus_i H^{2i}(S, \mu_S^{\otimes i}) \longrightarrow \bigoplus_i H^{2i}(P, \mu_P^{\otimes i}) = A^*(P)$$

is an injective morphism of unital rings, and the elements $1, \xi, \dots, \xi^r$ form a basis of the $A^*(S)$ -module $A^*(P)$. In particular, $A^*(P)$ is a free $A^*(S)$ -module of the same rank as E .

3. Chern Classes

Let S be a scheme and let E be a locally free $\mathcal{O}_S$ -module of constant rank r . Denoting by $\check{E}$ the dual of E , put $P = P_S(\check{E})$. We have seen (2.2.6) that, if ξ denotes the class of $\mathcal{O}_P(1)$ in $H^2(P, \mu)$, then the $A^*(S)$ -module $A^*(P)$ is free with basis $1, \xi, \dots, \xi^{r-1}$. In particular, it is immediate for reasons of homogeneity that there exists a unique relation of the form

$$\xi^r + c_1 \xi^{r-1} + \dots + c_i \xi^{r-i} + \dots + c_r = 0, \quad (3.1)$$

with $c_i \in H^{2i}(S, \mu^{\otimes i})$ for every $i \in [0, r]$.

Now let E be any locally free $\mathcal{O}_S$ -module. For every integer r , the set S_r of points of S at which E has rank r is open. Thus one obtains a partition

$$S = \coprod_r S_r$$

of S into disjoint opens; in particular, for every A_S -module M and every integer i , the product of the inverse-image morphisms

$$H^i(S, M) \longrightarrow \prod_r H^i(S_r, M/S_r)$$

is a bijection.

The preceding considerations justify the following definition.

Definition 3.2. Let S be a scheme and E a locally free $\mathcal{O}_S$ -module. For every integer $i \geq 0$, the i -th Chern class of E , denoted $c_i(E)$, is the element of $H^{2i}(S, \mu^{\otimes i})$ whose restriction, for every integer $r \geq 0$, to the open subset S_r of S consisting of the points where E has rank r , is the element c_i defined by relation 3.1. If the rank of E is bounded, the total Chern class of E is the element

$$c(E) = \sum_i c_i(E)$$

of $A^*(S)$; in the general case, this denotes the corresponding element of

$$\widehat{A}^*(S) = \prod_i A^i(S).$$

3.3

It is evident that, for all S and E , $c_0(E) = 1$, and that, if the rank of E is bounded by d , then $c_i(E) = 0$ for $i > d$.

Proposition 3.4. The Chern classes just defined satisfy the following properties.

- (i) *Normalization.* Let L be an invertible $\mathcal{O}_S$ -module. If $[L]$ denotes the image of the class of L under the boundary morphism

$$H^1(S, \mathbb{G}_m) \longrightarrow H^2(S, \mu),$$

then

$$c(L) = 1 + [L].$$

- (ii) *Stability under base change.* Let $f : S' \rightarrow S$ be a morphism of schemes and let E be a locally free $\mathcal{O}_S$ -module. For every integer i ,

$$f^*(c_i(E)) = c_i(f^*E).$$

(iii) *Additivity.* Let

$$0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$$

be an exact sequence of locally free $\mathcal{O}_S$ -modules. For every integer n , one has

$$c_n(E) = \sum_{i+j=n} c_i(E')c_j(E'').$$

Before beginning the proof, we extract the following corollary.

Corollary 3.5. For every exact sequence $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ of locally free $\mathcal{O}_S$ -modules, one has

$$c(E) = c(E')c(E'')$$

in $\hat{A}^*(S)$.

Proof of 3.4. Following A. Grothendieck: *La théorie des classes de Chern*, Bull. S.M.F. (1958), pp. 137–154.

(i) Let E be a locally free $\mathcal{O}_S$ -module; put $P = P_S(\check{E})$ and let $p : P \rightarrow S$ be the canonical projection. Recall that there is a canonical surjective homomorphism

$$p^*(\check{E}) \longrightarrow \mathcal{O}_P(1).$$

In particular, if E is invertible and we put $E = L$, in agreement with the notation of (i), then $P = S$ and there is an isomorphism

$$L \xrightarrow{\sim} \mathcal{O}_P(1).$$

Therefore $\xi + [L] = 0$, and the assertion follows from the definition of Chern classes.

(ii) Immediate verification, left to the reader.

(iii) By using a partition of S into opens on which the modules in question have constant rank, we may suppose that E, E', E'' have constant ranks r, r', r'' , respectively. Suppose that the following lemma has been proved; we show how it implies the assertion.

Lemma 3.6. Let E be a locally free $\mathcal{O}_S$ -module endowed with a finite filtration

$$0 = E_0 \subset E_1 \subset \cdots \subset E_i \subset \cdots \subset E_r = E$$

whose consecutive quotients $L_i = E_i/E_{i-1}$ ($1 \leq i \leq r$) are invertible. Then

$$c(E) = (1 + [L_1])(1 + [L_2]) \cdots (1 + [L_r]).$$

Let $h : Z \rightarrow S$ be the fibre product of the flag schemes of E' and E'' . By the definition of flag schemes, it is clear that $h^*(E')$ and $h^*(E'')$ admit finite filtrations with invertible quotients (L'_i) ($1 \leq i \leq r'$) and (L''_j) ($1 \leq j \leq r''$), respectively. It follows that $h^*(E)$ admits a filtration with consecutive quotients the L'_i and the L''_j . Applying Lemma 3.6 then shows that

$$c(h^*(E)) = \prod_{1 \leq i \leq r'} c(L'_i) \prod_{1 \leq j \leq r''} c(L''_j) = c(h^*(E'))c(h^*(E'')),$$

hence, by (ii),

$$h^*(c(E)) = h^*(c(E')c(E'')). \quad (3.5.1)$$

But h is a composite of morphisms of the form

$$P_T(F) \longrightarrow T \quad (F \text{ locally free } \mathcal{O}_T\text{-module}),$$

so by (2.2.5) the inverse-image homomorphism $h^* : A^*(S) \rightarrow A^*(Z)$ is injective. This gives the desired assertion by (3.5.1).

We prove Lemma 3.6. By the preceding argument, it is enough to prove the analogous relation

$$c(p^*E) = \prod_i c(p^*L_i),$$

where, recall, $p : P = P_S(\check{E}) \rightarrow S$ is the canonical projection. For every $i \in [1, r]$, the monomorphism $E_i \rightarrow E$ defines by duality an epimorphism $\check{E} \rightarrow \check{E}_i$; putting $Y_i = P_S(\check{E}_i)$, one obtains a sequence of closed subschemes of P :

$$S = Y_1 \subset Y_2 \subset \cdots \subset Y_i \subset \cdots \subset Y_r = P.$$

For every pair (i, j) of integers with $1 \leq i \leq j \leq r$, let

$$\alpha_{ji} : Y_i \rightarrow Y_j$$

be the canonical closed immersion, and put $\alpha_i = \alpha_{r,i}$. For every integer $i \in [1, r-1]$, it follows from the evident lemma (3.6.1) below that Y_i is defined in Y_{i+1} by an invertible ideal isomorphic to

$$\alpha_{i+1}^*(p^*(\check{L}_{i+1}) \otimes \mathcal{O}_P(-1)).$$

Lemma 3.6.1. Let $0 \rightarrow L \rightarrow E \rightarrow F \rightarrow 0$ be an exact sequence of locally free $\mathcal{O}_S$ -modules. Then the sequence

$$0 \rightarrow L \otimes_S S(E) \xrightarrow{u} S(E) \xrightarrow{v} S(F) \rightarrow 0,$$

where v denotes the evident morphism and u is the degree-one morphism corresponding to the inclusion of L in E , is an exact sequence of graded $\mathcal{O}_S$ -modules.

Moreover, Y_1 is identified with S and we have an isomorphism

$$\alpha_1^*(p^*(\check{L}_1)) \simeq \mathcal{O}_{Y_1}(1),$$

i.e.

$$\alpha_1^*(p^*(\check{L}_1) \otimes \mathcal{O}_P(-1)) \simeq \mathcal{O}_{Y_1}. \quad (3.6.2)$$

We shall be able to conclude by using the following two properties of the Gysin homomorphism (SGA 4 XVIII) associated with a closed immersion $u : Z \rightarrow X$ of smooth S -schemes defined by an ideal J of $\mathcal{O}_X$:

$$u_*(x u^*(y)) = u_*(x)y, \quad (3.6.3)$$

and, if the ideal J is invertible, the relation

$$u_*(1_Z) = -[J] = [\check{J}]. \quad (3.6.4)$$

Indeed, from this we shall prove, by induction increasing on the integer i , that

$$(\alpha_i)_*(\alpha_i)^*\left(\prod_{j \leq i} [p^*(\check{L}_j) \otimes \mathcal{O}_P(-1)]\right) = 0. \quad (3.6.5)$$

For $i = r$, assertion (3.6.5) says

$$\prod_i [p^*(\check{L}_i) \otimes \mathcal{O}_P(-1)] = 0,$$

or equivalently

$$\prod_i (\xi + c_1(p^*(\check{L}_i))) = 0,$$

which immediately gives (3.6). Assertion (3.6.5) for $i = 1$ follows from (3.6.2). Suppose it true for i and prove it for $i + 1$. For simplicity put

$$N = \prod_{j \leq i} [p^*(\check{L}_j) \otimes \mathcal{O}_P(-1)].$$

The equality (3.6.5) for i can also be written

$$(\alpha_{i+1})_*(\alpha_{i+1,i})_*(\alpha_{i+1,i})^*(\alpha_{i+1})^*(N) = 0,$$

hence, in view of the projection formula for $\alpha_{i+1,i}$,

$$(\alpha_{i+1})_*[(\alpha_{i+1})^*(N)(\alpha_{i+1})^*(p^*(\check{L}_{i+1}) \otimes \mathcal{O}_P(-1))] = 0,$$

which is exactly the desired result.

Corollary 3.7.

- (i) For every locally free $\mathcal{O}_S$ -module E and every integer i ,

$$c_i(\check{E}) = (-1)^i c_i(E).$$

- (ii) If E is a locally free $\mathcal{O}_S$ -module of rank r , then

$$c_1(E) = c_1(\Lambda^r E).$$

- (iii) More generally, the Chern classes of a tensor product or of an exterior power are expressed by means of the universal polynomials made explicit in (SGA 6 V 6.1).

Proof. We prove, for example, (ii); the other assertions are seen similarly. Using the flag scheme of E , one reduces to the case where E admits a filtration with invertible quotients L_i . The assertion then follows from the well-known isomorphism

$$\Lambda^r(E) \simeq \bigotimes_i L_i.$$

3.8

Given a scheme X locally of finite type over $\text{Spec}(\mathbb{C})$, the set $X(\mathbb{C})$ of its rational points is canonically endowed with the structure of a locally compact topological space. Thus, by algebraic topology, one has a theory of Chern classes [3], assigning to every continuous complex vector bundle V on $X(\mathbb{C})$ classes $c_i(V) \in H^{2i}(X(\mathbb{C}), \mathbb{Z})$. Now let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of continuous complex-valued functions on $X(\mathbb{C})$, and let $\mathcal{E}$ be a locally free $\mathcal{O}_{\mathbb{C}}$ -module of finite type. One defines Chern classes

$$c_i(\mathcal{E}) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}),$$

namely those of the complex vector bundle whose sheaf of continuous sections is $\mathcal{E}$. Finally, using the evident map $\mathbb{Z} \rightarrow \mathbb{Z}/\nu\mathbb{Z}$, one obtains Chern classes, denoted by the same symbol when no confusion is possible,

$$c_i(\mathcal{E}) \in H^i(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}). \quad (3.8.1)$$

On the other hand, when $X_{\text{ét}}$ is equipped with its structural sheaf $\mathcal{O}_a$ and $X(\mathbb{C})$ with the ring $\mathcal{O}_{\mathbb{C}}$, there is a morphism of ringed topoi (SGA 4 XI)

$$\varepsilon : X(\mathbb{C}) \rightarrow X_{\text{ét}},$$

the structural morphism of rings

$$\varepsilon^*(\mathcal{O}_a) \rightarrow \mathcal{O}_{\mathbb{C}} \quad (3.8.2)$$

coming from the fact that every algebraic function is continuous. The morphism ε induces a morphism of abelian sheaves

$$\varepsilon^*(\mu_{X_{\text{ét}}}) \rightarrow \mu_{X(\mathbb{C})}, \quad (3.8.3)$$

where the letter μ denotes, in each of the two contexts, the sheaf of ν -th roots of unity. Finally, we shall canonically identify $(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})}$ and $\mu_{X(\mathbb{C})}$ by means of the isomorphism

$$(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})} \xrightarrow{\sim} \mu_{X(\mathbb{C})} \quad (3.8.4)$$

sending the section 1 to the section $e^{2i\pi/\nu}$.

Proposition 3.8.5. Let E be a locally free $\mathcal{O}_X$ -module. For every integer $i \geq 0$, let

$$\varepsilon^* : H^{2i}(X_{\text{ét}}, \mu_X^{\otimes i}) \rightarrow H^{2i}(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z})$$

be the inverse-image morphism deduced from 3.8.3 and 3.8.4. Then

$$c_i(\varepsilon^*E) = \varepsilon^*c_i(E).$$

Proof. It is enough to check this when E is invertible. Then, with the usual identifications, $\varepsilon^*(E)$ is the image of E under the canonical map

$$\varepsilon^* : H^1(X_{\text{ét}}, \mathbb{G}_m) \rightarrow H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*), \quad (3.8.6)$$

and one has to see that the diagram

$$\begin{array}{ccc} H^1(X, \mathbb{G}_m) & \xrightarrow{\varepsilon^*} & H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*) \\ \partial \downarrow & & \downarrow c_1 \\ H^2(X, \mu_X) & \xrightarrow{\varepsilon^*} & H^2(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}) \end{array} \quad (3.8.7)$$

where ∂ is the boundary morphism coming from the canonical exact sequence

$$0 \rightarrow \mu_X \rightarrow \mathbb{G}_m \xrightarrow{\nu} \mathbb{G}_m \rightarrow 0 \quad (3.8.8)$$

and c_1 is the first Chern class (3.8.1), is commutative. In fact, using that the morphism "first Chern class"

$$H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*) \rightarrow H^2(X, \mathbb{Z})$$

is the boundary morphism of the canonical exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \rightarrow 0, \quad t \mapsto e^{2i\pi t},$$

one easily sees that the morphism c_1 of (3.8.7) is nothing other than the boundary morphism associated with the exact sequence

$$0 \rightarrow \mathbb{Z}/\nu\mathbb{Z} \rightarrow \mathcal{O}_{\mathbb{C}}^* \xrightarrow{\nu} \mathcal{O}_{\mathbb{C}}^* \rightarrow 0, \quad t \mapsto e^{2i\pi t/\nu},$$

which, in view of (3.8.4), is the analogue of (3.8.8). The assertion follows at once.

3.9

Let X be a scheme and E a locally free $\mathcal{O}_X$ -module. Immediate passage to the limit allows one to define, for every prime number ℓ prime to the residual characteristics of X , ℓ -adic Chern classes

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_{\ell}(i)) \stackrel{\text{def}}{=} \varprojlim_{\nu} H^{2i}(X, \mu_{\ell^{\nu}}^{\otimes i}),$$

which have the usual properties made explicit in 3.4 and 3.7 and which, in the context of 3.8, are manifestly compatible, in an evident sense, with the Chern classes defined topologically.

Tensoring by $\mathbb{Q}_{\ell}$, one obtains ℓ -adic Chern classes modulo torsion

$$c_i(E) \in H^{2i}(X, \mathbb{Q}_{\ell}(i)) = H^{2i}(X, \mathbb{Z}_{\ell}(i)) \otimes_{\mathbb{Z}_{\ell}} \mathbb{Q}_{\ell},$$

which manifestly satisfy the same properties.

Now suppose that X is a quasi-projective smooth scheme over a field k , and denote here by $C^*(X)$ its Chow ring (Sém. Chevalley: Anneaux de Chow et applications III 4). Recall that in IV we defined a morphism of graded rings

$$\varphi_X : C^*(X) \rightarrow A^*(X)$$

compatible with direct-image and inverse-image morphisms. Moreover, the canonical isomorphism

$$\varepsilon_X : \text{Pic}(X) \xrightarrow{\sim} C^1(X)$$

which sends every invertible $\mathcal{O}_X$ -module L to the class of the divisor of a rational section of L not vanishing on any connected component of X , gives, on the model of the aforementioned article of Grothendieck, where k was assumed algebraically closed, a theory of Chern classes with values in $C^*(X)$. The composite

$$\varphi_X \circ \varepsilon_X : \text{Pic}(X) \longrightarrow H^2(X, \mathbb{Z}_{\ell}(1))$$

is just the first ℓ -adic Chern class. Indeed, it is enough to check this on elements of $\text{Pic}(X)$ of the form $\check{J}$, with J an invertible ideal of $\mathcal{O}_X$, and then this follows from (3.6.4). By the uniqueness properties of theories of Chern classes, it follows that the morphism φ_X sends Chern classes to Chern classes.

4. Self-intersection Formula and Applications

Let S be a scheme, X an S -scheme, and Y a closed subscheme of X , smooth over S and everywhere of codimension d . Finally suppose that X is smooth over S in a neighbourhood of Y . Let $i : Y \rightarrow X$ be the canonical closed immersion and J the ideal of $\mathcal{O}_X$ defining Y . Recall (SGA 4^{1/2} Cycle) that in this case one has Gysin isomorphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \rightarrow H_Y^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

namely cup-products with the fundamental class. If no confusion is possible, we shall denote by the same symbol the Gysin morphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \rightarrow H^{2p+2d}(X, \mu_X^{\otimes(p+d)})$$

obtained from the preceding ones by forgetting supports.

Theorem 4.1. In the preceding situation, if

$$N = J/J^2$$

denotes the normal sheaf of i , then for every $y \in A^*(Y)$ one has

$$i^* i_*(y) = y c_d(\check{N}).$$

Before beginning the proof of this theorem, we indicate a few variants, proved in exactly the same way.

a) If ℓ is a prime number prime to the residual characteristics of S , the analogous formula is true in $H^*(Y, \mathbb{Z}_\ell(*))$.

b) One may put in supports. If, for example, T is a closed subset of Y and $y \in H_T^{2p}(Y, \mu_Y^{\otimes p})$, formula 4.1 is still true in $H_T^{2p+2d}(Y, \mu_Y^{\otimes(p+d)})$. Of course, i_* then denotes the Gysin morphism

$$H_T^{2p}(Y, \mu_Y^{\otimes p}) \rightarrow H_T^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

which is again an isomorphism (SGA 4^{1/2} Cycle).

c) Finally, the two cases may be combined.

Since the fundamental tool in the proof of 4.1 is the theory of blown-up schemes, we now recall a few facts about it. Let

$$f : X' \rightarrow X$$

be the X -scheme obtained by blowing up Y , and let

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ g \downarrow & & \downarrow f \\ Y & \xrightarrow{i} & X \end{array}$$

be the cartesian diagram constructed from f and i . Recall (cf. SGA 6 VII 3.1, whose notation we adopt) that j is a regular immersion of codimension 1 defined by an ideal

$$J' \simeq \mathcal{O}_{X'}(1)$$

and that $Y' = P_Y(N)$. Moreover, one has an exact sequence of $\mathcal{O}_{Y'}$ -modules

$$0 \rightarrow F \rightarrow g^*(N) \rightarrow \mathcal{O}_{Y'}(1) \rightarrow 0.$$

Lemma 4.2. For every A_X -module M , the homomorphisms

$$f^* : H_Y^*(X, M) \rightarrow H_{Y'}^*(X', f^*(M))$$

and

$$f_* : H_{Y'}^*(X', f^*(M)) \rightarrow H_Y^*(X, M)$$

are related by $f_* f^* = \text{id}$. Moreover, f_* has degree 0.

Proof. Since f is a complete-intersection morphism of relative dimension zero (SGA 6 VII 1.8), the last assertion follows from the definition of the Gysin homomorphism (SGA 4^{1/2} Cycle). For the first, the projection formula for f ,

$$f_*(x f^*(x')) = f_*(x) x',$$

where $x \in H^*(X, A)$ and $x' \in H_{Y'}^*(X, M)$, reduces the claim to proving that $f_*(1) = 1$. It is enough to check the analogous assertion over $U = X - Y$, which is evident because f induces an isomorphism from $U' = X' - Y'$ onto U .

Remark 4.3. Of course, the analogue of 4.2 obtained by eliminating supports is also true and is proved in the same way.

We now come to the proof of 4.1. First, by a general result of (SGA 4^{1/2} Cycle), which is proved in an essentially formal way, one has

$$i^* i_*(y) = y i^* i_*(1),$$

so it is enough to prove

$$i^* i_*(1_Y) = c_d(\check{N}),$$

where i_* here denotes the Gysin isomorphism

$$H^0(Y, A_Y) \rightarrow H_Y^{2d}(X, \mu_X^{\otimes d}).$$

Since $f^* i_* 1_Y \in H_{Y'}^{2d}(X', \mu_{X'}^{\otimes d})$, and since the Gysin morphism

$$j_* : H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)}) \rightarrow H_{Y'}^{2d}(X', \mu_{X'}^{\otimes d})$$

is an isomorphism, there exists

$$u \in H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)})$$

such that

$$f^* i_*(1_Y) = j_*(u). \tag{4.1.1}$$

Applying j^* to both sides of 4.1.1, one obtains

$$j^* f^*(1_Y) = j^* j_*(u).$$

Putting $\xi = c_1(\mathcal{O}_{Y'}(1))$, it follows from 3.6.4 that

$$g^* i^* i_*(1_Y) = -u \xi. \quad (4.1.2)$$

The element u admits a unique decomposition of the form

$$u = a_0 + a_1 \xi + \cdots + a_{d-1} \xi^{d-1}, \quad a_i \in H^{2d-2-2i}(Y, \mu_Y^{\otimes(d-1-i)}). \quad (4.1.3)$$

In view of the defining relation for the Chern classes of $\check{N}$,

$$\xi^d + c_1(\check{N}) \xi^{d-1} + \cdots + c_{d-1}(\check{N}) \xi + c_d(\check{N}) = 0, \quad (4.1.4)$$

we deduce the equality

$$u \xi = -a_{d-1} c_d(\check{N}) + (a_0 - a_{d-1} c_{d-1}(\check{N})) \xi + \cdots + (a_{d-2} - a_{d-1} c_1(\check{N})) \xi^{d-1}.$$

Comparing this last expression with (4.1.2), and using the injectivity of g^* (2.2.6), one obtains

$$i^* i_*(1_Y) = a_{d-1} c_d(\check{N}). \quad (4.1.5)$$

It remains to determine a_{d-1} . First, the homomorphism g_* lowers degrees by $2d-2$, hence

$$g_*(\xi^i) = 0 \quad (0 \leq i \leq d-2),$$

and moreover (2.2.2)

$$g_*(\xi^{d-1}) = 1.$$

Applying g_* to both sides of 4.1.3, we obtain

$$g_*(u) = a_{d-1}. \quad (4.1.6)$$

On the other hand, applying f_* to both sides of 4.1.1, we obtain, in view of 4.2,

$$i_*(1_Y) = i_* g_*(u). \quad (4.1.7)$$

But the homomorphism i_* in question is injective; hence

$$a_{d-1} = g_*(u) = 1_Y,$$

which completes the proof.

Remarks 4.4. a) The essential ingredient in what precedes is the purity theorem. Once it is available in full generality, statement 4.1 will be valid with Y regular and purely of codimension d in X , and X excellent and regular in a neighbourhood of Y . In this form, the result is in any case established in characteristic 0, by (SGA 4 XIX).

b) When S is the spectrum of a separably closed field and X and Y are quasi-projective and smooth over S , the analogous statement in the Chow ring of Y was established by Mumford (cf. 9.2).

Corollary 4.5. Under the hypotheses of 4.1, let x and y belong to $A^*(Y)$. Then

$$i_*(x)i_*(y) = i_*(xy c_d(\check{N})).$$

Proof. The projection formula for the morphism i shows that

$$i_*(x)i_*(y) = i_*(x i^* i_*(y)),$$

and the conclusion follows from 4.1.

Corollary 4.6. Let S be a scheme, let E be a locally free $\mathcal{O}_S$ -module of constant rank r , and let $g : X = V(\check{E}) \rightarrow S$ be the vector bundle whose sheaf of sections is E . For every section f of X over S , one has

$$f_*(1_S) = g^*(c_r(E))$$

in $H^{2r}(X, \mu_X^{\otimes r})$.

Proof. Since g^* is an isomorphism (1.2) and $g \circ f = \text{id}$, f^* is bijective. It is therefore enough to check that

$$f^* f_*(1_S) = f^* g^*(c_r(E)).$$

If N denotes the normal sheaf of f , this last equality is equivalent, by 4.1, to

$$c_r(\check{N}) = c_r(E).$$

In fact one even has an isomorphism $\check{N} \simeq E$, since the exact sequence of (EGA IV 17.2.5), applied to the situation

$$S \xrightarrow{f} X \xrightarrow{g} S,$$

gives an isomorphism $N = f^* \Omega_{X/S}^1$, and moreover $\Omega_{X/S}^1 \simeq g^* \check{E}$ (EGA IV 16.4.8).

Corollary 4.7. Under the hypotheses of 4.6, assume in addition that S is smooth and quasi-projective over a separably closed field. Let σ_X denote the zero-section cycle of X , and let $Y = f^{-1}(\sigma_X)$ be the cycle of zeros of f . Then

$$\text{cl}_S(Y) = c_r(E)$$

in $H^{2r}(S, \mu_S^{\otimes r})$.

Proof. By IV,

$$\text{cl}_S(Y) = f^* \text{cl}_X(\sigma_X).$$

Let s_0 be the zero-section morphism of X . Then

$$\text{cl}_X(\sigma_X) \stackrel{\text{dfn}}{=} (s_0)_*(1_S) \stackrel{(4.6)}{=} g^*(c_r(E)),$$

and the assertion follows by applying f^* to the extreme terms.

Remark 4.8. Let ℓ be a prime number distinct from the residual characteristics of S . By simple passage to the projective limit, one deduces from 4.6 and 4.7 the corresponding statements in ℓ -adic cohomology.

Now let ℓ be a prime number and let X be a proper scheme over a separably closed field k of characteristic prime to ℓ . Recall (IV) that the ℓ -adic Euler-Poincaré characteristic of X is the integer

$$\mathrm{EP}_\ell(X) = \sum_i (-1)^i \dim H^i(X, \mathbb{Q}_\ell).$$

Corollary 4.9. Let X be a smooth, proper, connected scheme of dimension n over a separably closed field k , with $\mathrm{char}(k) \neq \ell$. If $p : X \rightarrow k$ denotes the canonical projection, then

$$\mathrm{EP}_\ell(X) = p_* c_n(\check{\Omega}_X^1)$$

in $\mathbb{Q}_\ell = H^0(k, \mathbb{Q}_\ell)$.

Proof. Let $j : X \rightarrow X \times X$ be the diagonal immersion of X , and let $q : X \times X \rightarrow k$ be the canonical projection. Applying 4.1 to the morphism j gives

$$c_n(\check{\Omega}_X^1) = j^* j_*(1_X),$$

hence

$$p_*(c_n(\check{\Omega}_X^1)) = p_* j^* j_*(1_X) = q_* j_* j^* j_*(1_X).$$

Using the projection formula for j , one deduces

$$p_*(c_n(\check{\Omega}_X^1)) = q_*(j_*(1_X) \cup j_*(1_X)),$$

where, by IV, the second member is equal to the ℓ -adic Euler-Poincaré characteristic of X .

4.10

When X is projective over k , the result recalled from IV shows that $\mathrm{EP}_\ell(X)$ is, in the Chow ring of $X \times X$, the self-intersection of the diagonal cycle, and consequently is independent of the prime number $\ell \neq \mathrm{char}(k)$. We shall therefore be able to speak of the Euler-Poincaré characteristic $\mathrm{EP}(X)$ of X , without reference to the integer ℓ .

Proposition 4.11. Let X be a smooth, projective, connected scheme of dimension n over a separably closed field k . Then

$$\mathrm{EP}(X) = \sum_{p,q} (-1)^{p+q} \dim H^p(X, \Omega_X^q).$$

Proof. We use the notation of the proof of 4.9, and let J be the ideal of the diagonal immersion of X . As observed in 4.10, $\mathrm{EP}(X)$ is, in the Chow ring of $X \times X$, the intersection product $\Delta \cdot \Delta$ of the diagonal cycle Δ with itself. To compute it, one may neglect torsion and therefore work in $\mathrm{Gr}_{\mathrm{top}}^n(X)$. This shows that $\mathrm{EP}(X)$ is the image under the direct-image morphism

$$q_* : K(X \times X) \rightarrow K(k) = \mathbb{Z}$$

of the square of the class of $\mathcal{O}_{X \times X}/J$. But, by (SGA 6 VII 2.5), this square is equal to

$$\sum_i (-1)^i \lambda^i([\Omega_X^1]),$$

and the desired formula follows immediately, by the definition of q_* .

5. Flag Schemes

Let S be a scheme, let E be a locally free $\mathcal{O}_S$ -module of rank r , and let $(p_i)_{1 \leq i \leq m}$ be integers > 0 satisfying

$$p_1 + p_2 + \cdots + p_m = r.$$

Recall that a flag of type $(p_1, \dots, p_m)$ of E is a finite filtration

$$0 = E_m \subset E_{m-1} \subset \cdots \subset E_0 = E$$

of E such that, for every integer $i \in [1, m]$, the quotient E_{i-1}/E_i is locally free of rank p_i .

One defines a functor $(\text{Sch}/S)^\circ \rightarrow (\text{Ens})$ by assigning to an S -scheme $u : T \rightarrow S$ the set of flags of type $(p_1, \dots, p_m)$ of $u^*(E)$, and to a morphism of S -schemes the inverse image, in the evident sense, on flags. It is proved (EGA I, 2nd edition) that this functor is representable, and any representative is called an S -scheme of flags of type $(p_1, \dots, p_m)$ of E .

In what follows, $f : X \rightarrow S$ will denote the S -scheme of flags of type $(p_1, \dots, p_m)$ of E . Moreover, the scheme of flags of type $(1, 1, \dots, 1)$ of E will be denoted D and called the flag scheme of E , without further specification, if no confusion is possible.

It is clear from the definition that the $\mathcal{O}_X$ -module $f^*(E)$ admits a canonical finite filtration

$$0 = F_m \subset F_{m-1} \subset \cdots \subset F_0 = f^*(E),$$

for which the quotient $E_i = F_{i-1}/F_i$ ($1 \leq i \leq m$) is locally free of rank p_i . For every integer $i \in [1, m]$, let

$$g_i : D_i \rightarrow X$$

be the X -scheme of flags of E_i , and let $g : D_1 \times_X \cdots \times_X D_m \rightarrow X$ be the fibre product of the X -schemes D_i . It is clear that the composite

$$h = f \circ g : \prod_i D_i \rightarrow S \tag{5.1}$$

is S -isomorphic to the S -scheme of flags of E ; up to S -isomorphism, one may therefore write

$$D = \prod_i D_i.$$

Proposition 5.2. In $D^b(S, A)$ there is an isomorphism

$$\text{R}f_*(A) \simeq \bigoplus_i R^{2i}f_*(A)[-2i].$$

Proof. Since h is a composite of finitely many morphisms of the form

$$P_T(F) \rightarrow T \quad (F \text{ locally free } \mathcal{O}_T\text{-module}),$$

it follows from (2.2.1) that

$$\text{R}h_*(A) \simeq \bigoplus_j R^j h_*(A)[-j].$$

For the same reason, one also has

$$Rg_*(A) \simeq \bigoplus_j R^j g_*(A)[-j].$$

Using ([6] 2.16), one obtains an isomorphism

$$Rf_*(g_*A) \simeq \bigoplus_j R^j f_*(g_*A)[-j].$$

Since g is smooth with geometrically connected fibres, $A_X \simeq g_*(A_D)$; hence it remains to show that

$$R^{2i+1}f_*(A) = 0 \quad (i \in \mathbb{N}). \quad (5.2.1)$$

For this, one immediately reduces, using conservativity of the system of fibre functors (SGA 4 VIII 3.5), to the case where S is the spectrum of a separably closed field k . Then, since g is a composite of projective bundles, (2.2.4) shows that the inverse-image map

$$H^{2i+1}(X, A_X) \rightarrow H^{2i+1}(D, A_D)$$

is a monomorphism, so one may suppose that $X = D$. In this case, since the projection $D \rightarrow k$ is itself a composite of projective bundles, the assertion follows by induction from (2.2.4).

Corollary 5.3. For every morphism of schemes $u : S \rightarrow T$, the Leray spectral sequence

$$E_2^{pq} = R^p u_* R^q f_*(A) \Rightarrow R^{p+q}(u \circ f)_*(A)$$

is degenerate.

Proof. Cf. [6] 1.2.

We shall now specify the multiplicative structure of the cohomology of X , first examining a particular case. With the preceding notation, one obtains a morphism of rings

$$\alpha : A^*(S)[T_{ij_i}]_{1 \leq i \leq m, 1 \leq j_i \leq p_m} \rightarrow A^*(X)$$

by sending $A^*(S)$ into $A^*(X)$ by inverse image f^* and, for every pair (i, j) , as above, sending the element T_{ij} to the j -th Chern class $c_j(E_i)$. For every integer p , one has, by (3.4 (iii)),

$$c_p(f^*E) = \sum_{i_1+i_2+\dots+i_m=p} c_{i_1}(E_1) \cdots c_{i_m}(E_m),$$

so that, by functoriality of Chern classes, the morphism vanishes on the ideal J_X generated by the elements

$$c_p(E) - \sum_{i_1+\dots+i_m=p} T_{1,i_1} \cdots T_{m,i_m}$$

for varying p . Hence there is a morphism of rings

$$\varphi_X : (A^*(S)[T_{ij_i}]_{(i,j_i)})/J_X \rightarrow A^*(X).$$

Proposition 5.4. The above morphism of rings φ_X is an isomorphism. The $A^*(S)$ -module $A^*(X)$ is free of rank

$$\frac{r!}{p_1!p_2! \cdots p_m!}.$$

Proof. We follow Grothendieck's exposé (Séminaire Chevalley 1958, Anneaux de Chow et applications, Exp. IV). We shall need the following lemma (cf. Lemma 1, p. 19 of loc. cit.).

Lemma 5.4.1. a) Let R be a commutative ring, and let $(a_i)_{1 \leq i \leq r}$ be elements of R . Let σ_i be the i -th elementary symmetric function of the elements $(U_i)_{1 \leq i \leq r}$, and let I be the ideal of the polynomial ring $R[U_1, U_2, \dots, U_r]$ generated by the elements $\sigma_i - a_i$. Then the R -algebra

$$R[U_1, U_2, \dots, U_r]/I$$

is a free R -module of rank $r!$.

b) Let $r, p_1, p_2, \dots, p_m$ be integers > 0 satisfying $p_1 + \dots + p_m = r$. Then the morphism of polynomial rings

$$V = R[T_{i,j_i}]_{1 \leq i \leq m, 1 \leq j_i \leq p_i} \longrightarrow R[U_i]_{1 \leq i \leq r} = W$$

obtained by sending T_{i,j_i} to the j_i -th elementary symmetric function of the elements

$$(U_{p_1 + \dots + p_{i-1} + s})_{1 \leq s \leq p_i}$$

is injective and makes W a free V -module of rank $p_1! \dots p_m!$.

In particular, the morphism of R -algebras

$$R[T_1, \dots, T_r] \rightarrow R[U_1, \dots, U_r]$$

obtained by sending T_i to the i -th elementary symmetric function of the U_j is injective and makes the second member a free module of rank $r!$ over the first.

We show how (5.4.1) implies (5.4). First suppose that X is the scheme D of flags of E , and for simplicity put $T_{i1} = U_i$. Since the canonical projection $D \rightarrow S$ is a composite of $r - 1$ morphisms of the form

$$P_{Y_i}(M_i) \rightarrow Y_i \quad (1 \leq i \leq r - 1),$$

where M_i is a locally free $\mathcal{O}_{Y_i}$ -module of rank $r - i + 1$, it follows from (2.2.6) that $A^*(D)$ is a free $A^*(S)$ -module of rank $r!$, and that the morphism φ_D is surjective. But Lemma 5.4.1 a) shows that the source of φ_D is also a free $A^*(S)$ -module of rank $r!$; hence the assertion in the case $X = D$.

In the general case, it is clear that the diagram

$$\begin{array}{ccc} A^*(X) & \xrightarrow{g^*} & A^*(D) \\ \varphi_X \uparrow & & \uparrow \varphi_D \\ C = A^*(S)[T_{i,j_i}]/J_X & \xrightarrow{\gamma} & A^*(S)[U_k]/J_D \end{array}$$

commutes; here the map γ is obtained by passage to the quotient from the morphism of $A^*(S)$ -algebras

$$A^*(S)[T_{i,j_i}] \rightarrow A^*(S)[U_k]$$

defined by sending T_{i,j_i} to the j_i -th elementary symmetric function of the elements

$$(U_{p_1 + \dots + p_{i-1} + s})_{1 \leq s \leq p_i}.$$

Since φ_D is an isomorphism, 5.4.1 b) implies that $A^*(D)$ is a free C -module of rank $p_1! \cdots p_m!$. On the other hand, expressing g as a fibre product of X -schemes of flags shows that $A^*(D)$ is also a free $A^*(X)$ -module of rank $p_1! \cdots p_m!$. It follows at once that φ_X is an isomorphism.

In the statement of the next proposition, we agree to put, for every scheme Y and every A_Y -module M ,

$$Q(Y, M) = \bigoplus_{n,i} H^n(Y, M \otimes \mu_Y^{\otimes i}).$$

The A -module $Q(Y, M)$ is canonically endowed with a structure of $A^*(Y)$ -module, coming from cup-product. If $u : Z \rightarrow Y$ is a morphism of schemes, the inverse-image morphism

$$u^* : Q(Y, M) \rightarrow Q(Z, u^* M)$$

is compatible with the $A^*(Y)$ -module structure on $Q(Y, M)$ and the $A^*(Z)$ -module structure on $Q(Z, u^* M)$. One deduces a morphism of $A^*(Z)$ -modules, called canonical,

$$A^*(Z) \otimes_{A^*(Y)} Q(Y, M) \rightarrow Q(Z, u^* M).$$

Proposition 5.5. Let S be a scheme, and let E be a locally free $\mathcal{O}_S$ -module of rank r . Let $f : X \rightarrow S$ be the S -scheme of flags of type $(p_1, \dots, p_m)$ of E . Then, for every A_S -module L , the canonical morphism

$$\alpha : A^*(X) \otimes_{A^*(S)} Q(S, L) \rightarrow Q(X, f^* L)$$

is an isomorphism.

Proof. When $X = P_S(E)$, this is an immediate consequence of (2.2.4) and (2.2.6). Hence it follows that it is true for $X = D$. In the general case, since the canonical morphisms $g : D \rightarrow X$ and $h : D \rightarrow S$ are composites of projective bundles, the canonical morphisms

$$A^*(D) \otimes_{A^*(S)} Q(S, L) \rightarrow Q(D, h^* L)$$

and

$$A^*(D) \otimes_{A^*(X)} Q(X, g^* L) \rightarrow Q(D, g^* L)$$

are isomorphisms. Consequently $\text{id}_{A^*(D)} \otimes \alpha$ is an isomorphism, whence the result, since $A^*(D)$ is free over $A^*(X)$.

We shall now give a relative variant of (5.4) and (5.5). With the preceding notation, let $u : S \rightarrow T$ be a morphism of schemes. Put

$$A^*(X/T) = \bigoplus_i R^{2i}(u \circ f)_*(A_X)$$

and similarly

$$A^*(S/T) = \bigoplus_i R^{2i} u_*(A_S).$$

Finally, for every A_S -module L , put

$$Q(S/T, L) = \bigoplus_{n,i} R^{2i} u_*(L \otimes \mu_S^{\otimes i})$$

and

$$Q(X/T, f^* L) = \bigoplus_{n,i} R^n(u \circ f)_*(f^* L \otimes \mu_X^{\otimes i}).$$

For every pair (i, j) of integers, with $1 \leq i \leq m$ and $1 \leq j \leq p_i$, the image of the Chern class $c_j(E_i)$ by the canonical morphism

$$H^{2j}(X, \mu_X^{\otimes j}) \rightarrow H^0(T, R^{2j}(u \circ f)_*(\mu_X^{\otimes j}))$$

defines an A_T -morphism

$$c_{ij} : A_T \rightarrow R^{2j}(u \circ f)_*(\mu_X^{\otimes j}).$$

The data of the c_{ij} and the inverse image f^* allow one to define a morphism of $A^*(S/T)$ -algebras

$$\theta_1 : A^*(S/T)[T_{i,j_i}]_{(i,j_i)} \rightarrow A^*(X/T).$$

Denoting by c_i the sections of $A^*(S/T)$ defined by the Chern classes $c_i(E)$, one sees as in (5.4) that θ_1 vanishes on the ideal J locally generated by the relations

$$\sum_{i_1 + \dots + i_m = p} T_{1,i_1} \cdots T_{m,i_m} = c_p \quad (p \in \mathbb{N}).$$

Hence there is a morphism of $A^*(S/T)$ -algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \rightarrow A^*(X/T).$$

Proposition 5.6. With the preceding notation:

a) The morphism of $A^*(S/T)$ -algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \rightarrow A^*(X/T)$$

is an isomorphism.

b) For every A_S -module L , the canonical morphism of $A^*(X/T)$ -modules

$$A^*(X/T) \otimes_{A^*(S/T)} Q(S/T, L) \rightarrow Q(X/T, f^*L)$$

is an isomorphism.

Proof. Simple sheafification of (5.4) and (5.5), respectively.

6. Group Schemes

We begin with a general property of objects twisted by torsors.

Proposition 6.1. Let S be a scheme, G a quasi-compact smooth S -group scheme with connected fibres, and X an S -scheme endowed with a left action of G . If $f : E \rightarrow S$ is a right S -torsor under G , denote by X_E the S -scheme obtained from X by twisting by means of E and the action of G on X . Let $p : X \rightarrow S$ and $q : X_E \rightarrow S$ be the canonical projections. Under these conditions, for every A_S -module M and every integer i , the A_S -module $R^i q_*(q^*M)$ is canonically isomorphic to $R^i p_*(p^*M)$.

Proof.

$$\begin{array}{ccc}
E \times_S X & \xrightarrow{\varphi} & X_E \\
\text{pr}_1 \downarrow & & \downarrow q \\
E & \xrightarrow{f} & S
\end{array}
\quad
\begin{array}{ccc}
E \times_S X & \xrightarrow{\text{pr}_2} & X \\
\text{pr}_1 \downarrow & & \downarrow p \\
E & \xrightarrow{f} & S.
\end{array}$$

It is known that diagram (D_1) , where φ denotes the canonical passage-to-the-quotient map, is cartesian. Since f is smooth, the smooth base-change theorem gives an isomorphism

$$u : f^* R^i q_*(q^* M) \xrightarrow{\sim} R^i(\text{pr}_1)_*(\varphi^* q^* M).$$

Similarly, the cartesian diagram (D_2) gives an isomorphism

$$v : f^* R^i p_*(p^* M) \xrightarrow{\sim} R^i(\text{pr}_1)_*((\text{pr}_2)^* p^* M).$$

But $p \circ \text{pr}_2 = q \circ \varphi$, so after the evident identification of $\varphi^* q^*$ with $(\text{pr}_2)^* p^*$ one obtains an isomorphism

$$v^{-1} \circ u : f^* R^i q_*(q^* M) \xrightarrow{\sim} f^* R^i p_*(p^* M).$$

Since the geometric fibres of f are non-empty and connected, it follows from (SGA 4 XII 6.5 (i)) that the adjunction morphism

$$\text{adj} : \text{id} \rightarrow f_* f^*$$

is an isomorphism, and therefore

$$(\text{adj})^{-1} f_*(v^{-1} \circ u) \circ (\text{adj})$$

is the desired isomorphism.

Proposition 6.2. (Mme Raynaud [5].) Let S be a scheme and let $p : G \rightarrow S$ be a reductive S -group scheme (SGA 3 XIX). For every locally constant constructible A_S -module M and every integer i , the A_S -module $R^i p_*(p^* M)$ is locally constant and constructible. Moreover, for every morphism $u : S' \rightarrow S$ defining a cartesian diagram

$$\begin{array}{ccc}
G' & \xrightarrow{u'} & G \\
p' \downarrow & & \downarrow p \\
S' & \xrightarrow{u} & S,
\end{array}$$

the base-change morphism

$$u^* R^i p_*(p^* M) \rightarrow R^i p'_*(p \circ u')^*(M)$$

is an isomorphism.

Proof. The two assertions of the statement are local for the étale topology of S . By (SGA 3 XII 2.3), we may therefore suppose that G is split. Let B be a Borel subgroup of G of the type indicated in (SGA 3 XXII 5.1.1), so that the quotient G/B exists by (SGA 3 XXII 5.8.5), and let $q : G \rightarrow G/B$ and $r : G/B \rightarrow S$ be the canonical morphisms. Since r is proper and smooth, it satisfies the proper base-change theorem and the functors $R^i r_*$ send locally constant constructible abelian sheaves to sheaves of the same type (SGA 4 XVI 2.2). Moreover, by (SGA 4 XXII 5.5.1), the

S -scheme B is isomorphic to a finite product of vector bundles and punctured vector bundles, and consequently it follows immediately from (1.1) and (1.2) that, if $t : B \rightarrow S$ denotes the canonical morphism, the sheaves $R^i t_*(t^* M)$, with M a locally constant constructible A_S -module, are locally constant constructible and stable under base change. Now let M be a locally constant constructible A_S -module. The preceding considerations and (6.1) imply that the sheaves $R^i q_*(p^* M)$ are locally constant constructible; using the Leray spectral sequence

$$R^i r_*(R^j q_*(p^* M)) \Rightarrow R^{i+j} p_*(p^* M),$$

one deduces the same for the sheaves $R^i p_*(p^* M)$. Finally, by (6.1), the sheaves $R^i q_*(p^* M)$ are stable under base change; using the proper base-change theorem for r , it follows that the sheaves $R^i p_*(p^* M)$ are also stable under base change.

Remark 6.3. Since every reductive S -group is locally, for the étale topology, an inverse image by the canonical morphism $S \rightarrow \mathrm{Spec}(\mathbb{Z})$ of a reductive $\mathrm{Spec}(\mathbb{Z})$ -scheme (SGA 3 XXIII 5.7 and XXV 1.3), (6.2) shows that the computation of the cohomology, with coefficients of torsion prime to the characteristic, of reductive group schemes over an algebraically closed field reduces to the transcendental problem of computing the cohomology of reductive group schemes of the same type over the field of complex numbers; this is known by the methods of algebraic topology, in view of the comparison theorem (SGA 4 XI 4.4).

Corollary 6.4. Let S be a scheme and let $p : G \rightarrow S$ be an S -group scheme which is an extension of a finite étale S -group scheme F by a reductive S -group scheme H . Then the conclusions of (6.2) are valid for every locally constant constructible A_S -module M .

Proof. After the base change $F \rightarrow S$, we may suppose that F is of the form Γ_S , where Γ is an ordinary finite group. Then there is an isomorphism of S -schemes

$$G \simeq \coprod_{\gamma \in \Gamma} H,$$

and the conclusion follows from (6.2).

Corollary 6.5. Let S be a scheme, G a reductive S -group scheme, and H a reductive sub- S -group scheme of G , such that the quotient sheaf of G by H for the faithfully flat quasi-compact topology is representable. Then, if

$$p : G/H \rightarrow S$$

is the canonical morphism, the conclusions of (6.2) are valid for every locally constant constructible A_S -module M .

Proof. Let $f : G \rightarrow G/H$ be the canonical morphism; it is smooth, hence of descent for étale sheaves (SGA 4 VIII 9.1). For every integer $r \geq 0$, put

$$G_r = G \times_{G/H} G \times_{G/H} \cdots \times_{G/H} G \quad (r+1 \text{ times}),$$

which, by SGA 3 V 1.5, is canonically identified with

$$G \times_S H \times_S \cdots \times_S H,$$

where H occurs r times, and let $f_r : G_r \rightarrow G/H$ and $g_r : G_r \rightarrow S$ be the canonical morphisms, so that in particular $f_0 = f$. Finally, for every integer $q \geq 0$, let $H^q(M)$ denote the complex

$$R^q(g_0)_*(f_0^*M) \rightarrow R^q(g_1)_*(f_1^*M) \rightrightarrows \cdots \rightarrow R^q(g_r)_*(f_r^*M) \cdots$$

By (SGA), one has a spectral sequence of descent, or Čech spectral sequence,

$$E(M) : E_2^{pq} = H^p(H^q(M)) \Rightarrow R^{p+q}p_*(M).$$

Since the schemes G_r are reductive group schemes, it follows from 6.2 that the terms E_2^{pq} are locally constant constructible and compatible with every base change. It follows at once that the same is true for the E^n .

7. Complete Intersections

In this paragraph a separably closed field k is given. If ℓ is a prime number $\neq \text{char}(k)$, we denote, for every proper k -scheme X and every integer $i \geq 0$, by

$$b_\ell^i(X) \quad \text{or} \quad b^i(X)$$

when no confusion is possible, the i -th Betti number of X for coefficients $\mathbb{Z}_\ell$. If X is projective over k , a hyperplane section of X means the zero locus of a section transverse to the zero section of a vector bundle $V(\check{\mathcal{L}})$, with $\mathcal{L}$ an ample invertible $\mathcal{O}_X$ -module.

Theorem 7.1. (Lefschetz theorem.) Let X be a smooth, connected, projective k -scheme of dimension n , let Y be a subscheme of X which is the intersection of d hyperplane sections of X , and let L be a locally constant constructible A_X -module. Then:

(i) The inverse-image homomorphism

$$H^i(X, L) \rightarrow H^i(Y, L|Y)$$

is an isomorphism for $i \leq n - d - 1$ and a monomorphism for $i = n - d$.

(ii) Suppose Y is smooth over k , of dimension $n - d$. Then the Gysin homomorphism

$$H^i(Y, L|Y) \otimes \mu_Y^{\otimes d} \rightarrow H^{i+2d}(X, L)$$

is an isomorphism for $i \geq n - d + 1$ and an epimorphism for $i = n - d$.

Proof. It is well known that the complement of a hyperplane section is an affine open; consequently $U = X - Y$ is a union of d affine opens. Assertion (i) then follows immediately from the exact cohomology sequence

$$\cdots \rightarrow H_i^i(U, L|U) \rightarrow H^i(X, L) \rightarrow H^i(Y, L|Y) \xrightarrow{\partial} H_{i+1}^{i+1}(U, L|U) \rightarrow \cdots$$

and from the following lemma.